Lecture 1.8 – Lists and Data Frames

Specific Learning Objectives:

1.1.10 – Create vectors, arrays, matrices, lists, and data frames.

1.1.11 – Understand vectors and vectorized calculations.

1.1.12 – Learn how to index vectors, arrays, matrices, lists, and data frames.

Deep Thoughts on Gen AI: https://www.instagram.com/p/DOdnXJNkdtD/

Check Your Understanding

Which class of object would you use if you needed:

	List	Data frame
a) Members of different sizes	✓	✗
b) Members of different classes	✓	✓
c) Both a and b	✓	✗

Check Your Understanding

Create a list in which each member contains one of each data types you’ve learned so far in the course!

Check Your Understanding

In the `ToothGrowth` data set, how can you print out all the measured tooth lengths from their study?

How can you find the mean and standard deviation of these lengths?

Check Your Understanding

In the `ToothGrowth` data set, how can you print out all the measured tooth lengths from their study that were only given a dose of 1.0?

How can you find the mean and standard deviation of these lengths?

Can you find the mean of the tooth lengths for animals given an OJ dose of 1.0?

Using GenAI – have healthy skepticism

Load the Harman23.cor data set into R using:

```
> data("Harman23.cor")
```

Try this:

```
> mean(Harman23.cor$forearm)
```

It returns NA, but isn't suppose to be. The answer is 0.596.

Work with a partner and ask ChatGPT how to fix this. Try to figure out the right line of code that returns the correct answer.

Using GenAI – have healthy skepticism

Were you all able to figure it out?

How difficult was it?

Once you have the relevant data, you can compute the mean of the `forearm` values:

r   Copy code

```r
mean(forearm_data, na.rm = TRUE)
```

Important: Covariance Matrix Context

It's important to note that what you're extracting is not the raw data for `forearm`, but its covariance with other variables like `height`, `arm.span`, etc. If you are actually looking for the raw data of `forearm`, you would need the original dataset, not the covariance matrix.

If you're only working with the covariance matrix and still want to get something useful from it, consider that

A network error occurred. Please check your connection and try again. If this issue persists please contact us through our help center at help.openai.com.

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Regenerate

In-class Exercises

1. Catchup with assignments. Any questions on these?

2. Assignment 1.10

3. Assignment 1.11

Action Items

1. Complete Assignments 1.10 and 1.11.

2. Read Davies Ch. 6 for next time.